



Cloud Computing

Do you know that you can easily access your data any time on any device? Cloud storage makes this possible.



The Cloud is a server where large amounts of data can be stored. Cloud servers can be located anywhere in the world. We can access the cloud via the internet from anywhere and from any smart device.



Note:

The internet is a network of computers connected to each other.

Why do we need cloud computing?

The devices we use in our homes, schools or offices have limited storage space. To store more data, we require multiple hard drives. Our computers cannot run all software at once due to processing and storage limitations. Cloud computing can solve these problems. Through cloud computing, we can share software and storage space in nominal costs.

When do we use clouds?

We use a cloud during the following activities:

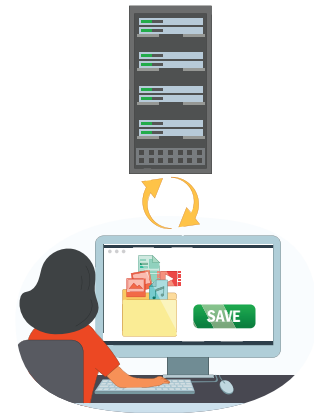
- a. Watching YouTube videos
- b. Playing online games
- c. Learning on an online platform

In short, most things that we do online are a part of cloud computing. Let us look at the features of cloud computing in more detail.

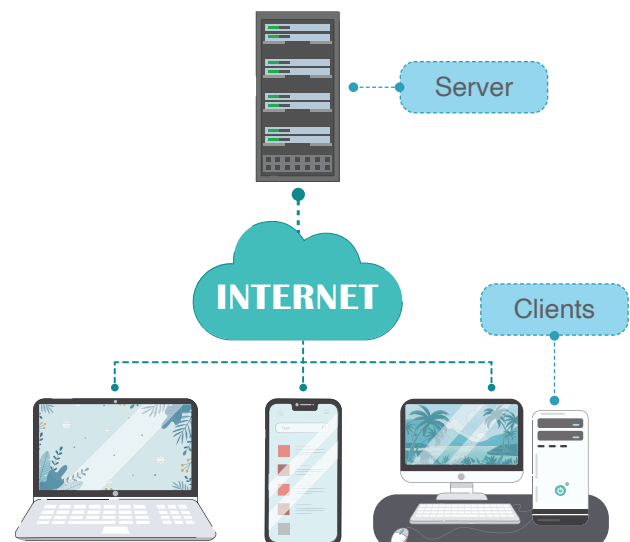
Storage

Nikki has a huge collection of songs, poems, and games. But her computer storage has no more space to store all of the data. What can she do?

Nikki can create an account on a cloud service and save all her data there.



To store and keep data safe on the cloud, a user has to create an account with a username and password. A server is a large storage device. The communication between a server and a user is called a client server system. The server is the computer which provides a service or sends information to the client. The client computer is the one which requests information.

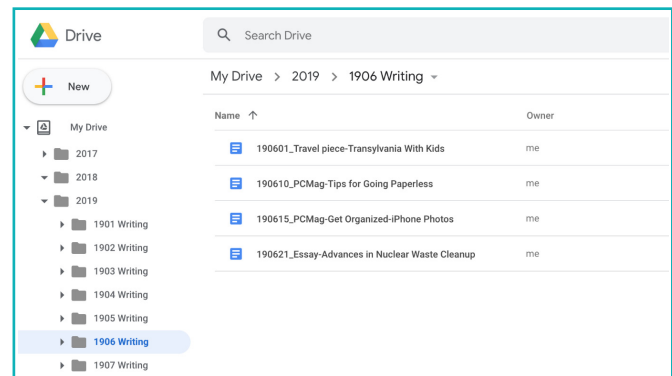


Examples of cloud storage applications:

Google Drive

Google Drive is one of the most popular cloud storage service. It can be used through desktop computers, mobiles, and tablets. Its structure is similar to the folder structure in computers.

We can create folders, share folders, as well as invite other users to collaborate on our files.



Let's see how easy it is to store data on the cloud through an activity.



Activity

- Discuss with your classmates and note down the websites or mobile applications you have used which require an internet connection.

Exercise

■ State whether True or False

- 1 Cloud has very little storage space.
▲ _____
- 2 To store and keep data safe on the cloud, a user has to create an account and password.
▲ _____
- 3 The server is a computer which provides a service or sends information.
▲ _____

■ Answer in one line

- 4 What is the cloud?
▲ _____

Service

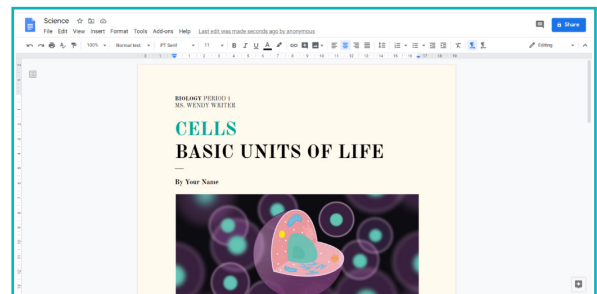
A cloud service provides computer software that can be used online. It is either a web-based or mobile application. When we use a cloud, we do not need to separately install the software on our computer. All of it is taken care of by the online software. It looks and acts exactly like the installed one.

Examples of cloud service applications

1. Google's cloud applications

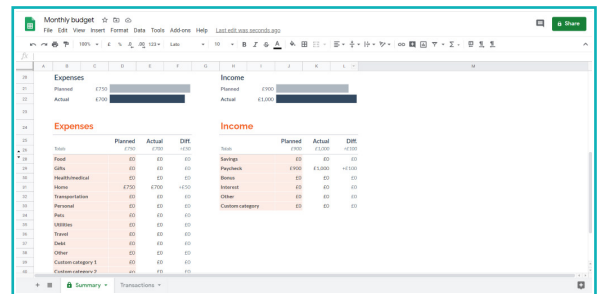
a. Google Docs

Google Docs is similar to MS Word that we use on our computers. We can create, save and share documents online.



b. Google Sheets

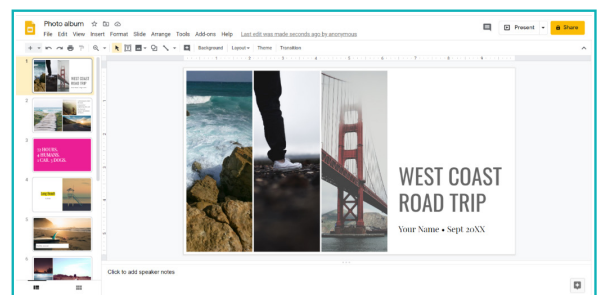
Google Sheets is similar to MS Excel or any other spreadsheet software that we use on our computers.



Expenses				Income			
	Planned	Actual	DIF		Planned	Actual	DIF
Food	£750	£760	£10	Salary	£1,000	£1,000	£0
Rent	£60	£60	£0	Payments	£1,000	£1,000	£0
Gas	£50	£50	£0	Bonus	£50	£50	£0
Healthcare	£50	£50	£0	Interest	£50	£50	£0
Home	£750	£750	£0	Other	£50	£50	£0
Transportation	£50	£50	£0	Custom category	£50	£50	£0
Personal	£50	£50	£0				
Phone	£50	£50	£0				
Utilities	£50	£50	£0				
Travel	£50	£50	£0				
Dining	£50	£50	£0				
Other	£50	£50	£0				
Custom category 1	£50	£50	£0				
Custom category 2	£50	£50	£0				

c. Google Slides

Google Slides is used to create and share presentations online.



2. Online games

These games can be played using the internet with real players and not just the computer.

3. Online shopping and food ordering

This kind of service allows us to shop online and order food online as well.

4. Messaging and calling

This type of service is used to message or call anyone online. It also provides audio and video call facilities.

Activity

Complete the following table with online and offline applications which provide similar features.

Desktop Software	Cloud Service
Word	Google Docs
Excel	
PowerPoint	
Files and Folders	

Features and advantages of cloud computing

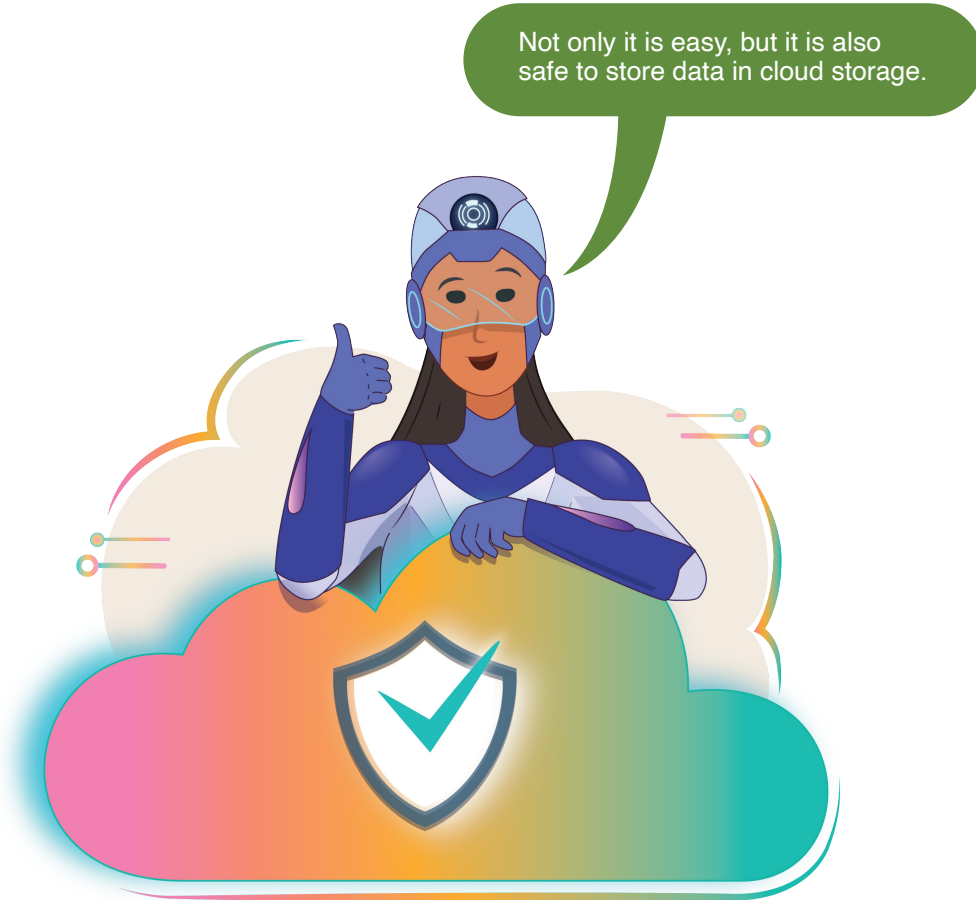
- a. Data from all over the world can be stored in a single location.
- b. The service can be used at any time.
- c. It costs less than purchasing new storage devices.
- d. Once online, a service can be accessed by people all over the world.
- e. The data is kept safe with individual login ids and passwords.

Cloud computing has managed to solve all of our data storage problems. It also provides facilities like shopping, playing games, and creating content online. It is cost effective, making it accessible for everyone. We can access our data using any device and from anywhere using cloud computing.

Exercise

Match the following:

- 5 a. Server
- b. Cloud Storage
- c. Client
- d. Cloud service
- e. Provides features of computer software online
- f. Requests information or a service
- g. Provides information and services to a computer requesting it
- h. Provides space to store data online



Tech Flash

- **Cloud Computing** : The cloud is a place where large amounts of data can be stored.
- **Server** : A server is a computer that provides services or information to the computer requesting it.
- **Client** : The client is a computer asking for a service or information from a server.
- **Cloud Storage** : Cloud storage is used to store data on servers, similar to data storage on a computer.
- **Cloud Service** : Cloud services provide the same features of a computer software online.
- **Web application** : A web application is a service which uses a web browser as an interface for online software.
- **Mobile application** : A mobile application is a software installed in smartphones.